Unit 3 :: Complex Analysis

☙ PHY0400404: Mathematical Physics ❧

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Syllabus

Unit 3: Complex analysis

Functions of complex variables, analyticity and Cauchy-Riemann conditions, examples of analytic functions; singular functions: poles and branch points, order of singularity; integration of functions with complex variable, Cauchy's integral theorem and Cauchy's integral formula, simply and multiply connected regions; Laurent and Taylor's series expansions, residue theorem with application.

Complex trigonometric and hyperbolic identities

1. $\cosh x = \dfrac{e^x + e^{-x}}{2}$
2. $\sinh x = \dfrac{e^x - e^{-x}}{2}$
3. $\cos x = \dfrac{e^{ix} + e^{-ix}}{2}$
4. $\sin x = \dfrac{e^{ix} - e^{-ix}}{2i}$

Using the above identities, it can be shown that

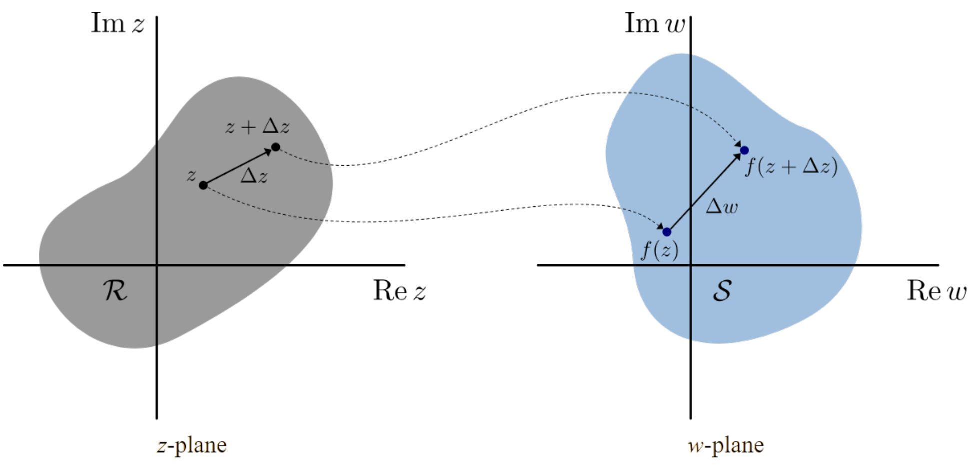
1. $\cosh ix = \cos x$
2. $\cos ix = \cosh x$
3. $\sinh ix = i \sin x$
4. $\sin ix = i \sinh x$

1 Functions of complex variables

Suppose, to each value that a complex variable z can assume, there corresponds one or more values of a complex variable w. We then say that w is a function of z and write $w = f(z)$.

- The complex variable z is called an *independent variable*
- The complex variable w is called a *dependent variable*.

1.1 Complex derivatives



If a complex function $f(z)$ is single-valued in some region $\mathcal{R}$ of the z plane, the derivative of $f(z)$ is defined as

$$f'(z) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

provided that the limit exists independent of the manner in which Δz approaches 0. In such a case, we say that $f(z)$ is *differentiable* at z and $f'(z)$ is known as the *derivative* of the function $f(z)$.

1.2 Complex analytic functions

If the derivative $f'(z)$ exists at all points z of a region $\mathcal{R}$, then $f(z)$ is said to be analytic in $\mathcal{R}$ and is referred to as an *analytic function* in $\mathcal{R}$.

A function $f(z)$ is said to be analytic at a point z_0 if there exists a neighborhood $|z - z_0| < \delta$ at all points of which $f'(z)$ exists.

Note: Analytic functions are sometimes also known as *regular* or *holomorphic* or *monogenic* functions.

1.3 Cauchy-Riemann conditions

Given a complex function which can be written as follows:

$$w = f(z) = u(x,y) + iv(x,y)$$

and is analytic over a region $\mathcal{R}$ on the z-plane, then the 2D functions $u(x,y)$ and $v(x,y)$ satisfy the following Cauchy-Riemann conditions in $\mathcal{R}$:

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}}$$

If the partial derivatives of u and v are continuous in $\mathcal{R}$, then the Cauchy-Riemann equations are sufficient conditions that $f(z)$ be analytic in $\mathcal{R}$.

Note: The functions $u(x,y)$ and $v(x,y)$ are known as *conjugate functions*.

Theorem 1

The necessary conditions for a function $f(z) = u(x,y) + iv(x,y)$ to be analytic at all points in a region $\mathcal{R}$ are

a) $\dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y}$

b) $\dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x}$

provided that $\dfrac{\partial u}{\partial x}, \dfrac{\partial u}{\partial y}, \dfrac{\partial v}{\partial x}$ and $\dfrac{\partial v}{\partial y}$ exist.

Theorem 2

The sufficient conditions for a function $f(z) = u(x,y) + iv(x,y)$ to be analytic at all points in a region $\mathcal{R}$ are

a) $\dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y}$ and $\dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x}$

b) $\dfrac{\partial u}{\partial x}, \dfrac{\partial u}{\partial y}, \dfrac{\partial v}{\partial x}$ and $\dfrac{\partial v}{\partial y}$ are continuous functions of x and y in the region $\mathcal{R}$.

Theorem 3

Given a function $f(z) = u(x,y) + iv(x,y)$, which is analytic over a region $\mathcal{R}$, its derivative is obtained as

$$f'(z) = \frac{\mathrm{d}f}{\mathrm{d}z} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

1.4 Examples of analyticity of complex functions

1. Determine whether $f(z) = \dfrac{1}{z}$ is analytic or not.
   Solution: Let us consider

$$\begin{aligned}
f(z) &= \frac{1}{z} \\
&= \frac{1}{x+iy} \\
&= \left(\frac{1}{x+iy}\right) \times \left(\frac{x-iy}{x-iy}\right) \\
&= \frac{x-iy}{(x+iy)(x-iy)} \\
&= \frac{x-iy}{x^2+y^2} \\
\implies f(z) &= \left(\frac{x}{x^2+y^2}\right) + i\left(\frac{-y}{x^2+y^2}\right) \\
&= u(x,y) + iv(x,y)
\end{aligned}$$

   This gives us

$$u(x,y) = \frac{x}{x^2+y^2} \quad \text{and} \quad v(x,y) = \frac{-y}{x^2+y^2}$$

   We obtain the partial derivatives as

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{(x^2+y^2)\times 1 - x \times 2x}{(x^2+y^2)^2} \\
&= \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} \\
&= \frac{y^2-x^2}{(x^2+y^2)^2}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial u}{\partial y} &= \frac{(x^2+y^2)\times 0 - x \times 2y}{(x^2+y^2)^2} \\
&= \frac{-2xy}{(x^2+y^2)^2}
\end{aligned}$$

$$\frac{\partial v}{\partial x} = \frac{(x^2+y^2)\times 0 - (-y)\times 2x}{(x^2+y^2)^2} = \frac{2xy}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{(x^2+y^2)\times(-1) - (-y)\times 2y}{(x^2+y^2)^2} = \frac{2y^2-x^2-y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

From the above partial derivatives, we find that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence the Cauchy-Riemann conditions are satisfied. Also the partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous everywhere except at $(0,0)$.

Therefore the complex function $f(z) = \frac{1}{z}$ is analytic everywhere except at $z = 0$.

2. Show that the function $f(z) = e^x(\cos y + i\sin y)$ is analytic, and then find its derivative.
Solution: Let us consider

$$\begin{aligned} & f(z) = e^x(\cos y + i\sin y) \\ \Longrightarrow\ & f(z) = (e^x\cos y) + i(e^x \sin y) \\ & \quad\ = u(x,y) + iv(x,y) \end{aligned}$$

This gives us

$$u(x,y) = e^x\cos y \quad \text{and} \quad v(x,y) = e^x \sin y$$

We obtain the partial derivatives as

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(e^x\cos y) = e^x\cos y$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(e^x\cos y) = -e^x\sin y$$

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x}(e^x\sin y) = e^x\sin y$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y}(e^x\sin y) = e^x\cos y$$

From the above partial derivatives, we find that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence the Cauchy-Riemann conditions are satisfied. Also the partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous everywhere.

Therefore the complex function $f(z) = e^x(\cos y + i \sin y)$ is analytic everywhere.

Finally, the derivative of the given function is obtained as

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \\ &= e^x \cos y + ie^x \sin y \\ \implies f'(z) &= e^x(\cos y + i \sin y) \end{aligned}$$

3. Test the analyticity of the function $w = \sin z$ and thereby show that $\dfrac{\mathrm{d}}{\mathrm{d}z}(\sin z) = \cos z$.

Solution: We have

$$\begin{aligned} w &= \sin z \\ &= \sin(x + iy) \\ &= \sin x \cos iy + \cos x \sin iy \\ &= \sin x(\cosh y) + \cos x(i \sinh y) \\ \implies w = f(z) &= \sin x \cosh y + i \cos x \sinh y \end{aligned}$$

This gives us

$$u(x, y) = \sin x \cosh y \quad \text{and} \quad v(x, y) = \cos x \sinh y$$

The partial derivatives are

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(\sin x \cosh y) \\ &= \cos x \cosh y \end{aligned} \qquad \begin{aligned} \frac{\partial u}{\partial y} &= \frac{\partial}{\partial y}(\sin x \cosh y) \\ &= \sin x \sinh y \end{aligned}$$

$$\begin{aligned} \frac{\partial v}{\partial x} &= \frac{\partial}{\partial x}(\cos x \sinh y) \\ &= -\sin x \sinh y \end{aligned} \qquad \begin{aligned} \frac{\partial v}{\partial y} &= \frac{\partial}{\partial y}(\cos x \sinh y) \\ &= \cos x \cosh y \end{aligned}$$

From the above partial derivatives, we find that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence the Cauchy-Riemann conditions are satisfied. Also the partial derivatives $\dfrac{\partial u}{\partial x}$, $\dfrac{\partial u}{\partial y}$, $\dfrac{\partial v}{\partial x}$ and $\dfrac{\partial v}{\partial y}$ are continuous everywhere.

Therefore the complex function $f(z) = \sin z$ is analytic everywhere.

Now, we evaluate the derivative as follows

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}z}(\sin z) &= \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \\ &= \cos x \cosh y + i(-\sin x \sinh y) \\ &= \cos x \cosh y - \sin x(i \sinh y) \\ &= \cos x(\cos iy) - \sin x(\sin iy) \\ &= \cos(x + iy) \\ \implies \frac{\mathrm{d}}{\mathrm{d}z}(\sin z) &= \cos z \end{aligned}$$

4. Find the values of C_1 and C_2 such that the function

$$f(z) = x^2 + C_1 y^2 - 2xy + i\big(C_2 x^2 - y^2 + 2xy\big)$$

is analytic. Also show that $f'(z) = 2(1+i)z$.

Solution: The given function is

$$f(z) = x^2 + C_1 y^2 - 2xy + i\big(C_2 x^2 - y^2 + 2xy\big)$$

from which we obtain

$$u(x,y) = x^2 + C_1 y^2 - 2xy \quad \text{and} \quad v(x,y) = C_2 x^2 - y^2 + 2xy$$

For the given function to be analytic, it should satisfy the Cauchy-Riemann conditions, i.e.

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \implies \frac{\partial}{\partial x}\big(x^2 + C_1 y^2 - 2xy\big) &= \frac{\partial}{\partial y}\big(C_2 x^2 - y^2 + 2xy\big) \\ \implies 2x - 2y &= -2y + 2x \end{aligned}$$

and

$$\begin{aligned} \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \\ \implies \frac{\partial}{\partial y}\big(x^2 + C_1 y^2 - 2xy\big) &= -\frac{\partial}{\partial x}\big(C_2 x^2 - y^2 + 2xy\big) \\ \implies 2C_1 y - 2x &= -(2C_2 x + 2y) \\ \implies 2C_1 y - 2x &= -2y - 2C_2 x \end{aligned}$$

Comparing the coefficients of y and x above, we obtain

$$\begin{aligned} 2C_1 = -2 \quad &\text{and} \quad -2C_2 = -2 \\ \implies C_1 = -1 \quad &\text{and} \quad C_2 = 1 \end{aligned}$$

Hence the given function is

$$f(z) = x^2 - y^2 - 2xy + i\big(x^2 - y^2 + 2xy\big)$$

Now, we have

$$\begin{aligned} \frac{\partial u}{\partial y} &= 2C_1 y - 2x \\ \implies \frac{\partial u}{\partial y} &= -2y - 2x \quad \text{and} \quad \frac{\partial v}{\partial y} = -2y + 2x \end{aligned}$$

Therefore, the derivative is

$$\begin{aligned} f'(z) &= \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y} \\ \implies f'(z) &= -2y + 2x - i(-2y - 2x) \\ &= -2y + 2x + 2i(x+y) \\ &= 2[(x + ix) + (-y + iy)] \\ &= 2[(1+i)x + i(1+i)y] \\ &= 2(1+i)(x+iy) \\ \implies f'(z) &= 2(1+i)z \end{aligned}$$

5. Obtain the Cauchy-Riemann conditions in polar form.
Solution: The polar representation of a complex number is

$$z = x + iy = re^{i\theta} \tag{1}$$

Differentiating equation (1) with respect to r and θ give us

$$\frac{\partial z}{\partial r} = e^{i\theta} \tag{2}$$

$$\frac{\partial z}{\partial \theta} = ire^{i\theta} \tag{3}$$

Let us now consider an analytic function given as

$$f(z) = u + iv \tag{4}$$

Differentiating equation (4) with respect to r gives us

$$\begin{aligned}
\frac{\partial}{\partial r}[f(z)] &= \frac{\partial}{\partial r}(u + iv) \\
\implies \frac{\mathrm{d}f}{\mathrm{d}z}\frac{\partial z}{\partial r} &= \frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r} \\
\implies f'(z)e^{i\theta} &= \frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r} \quad \text{[using eqn. (2) on LHS]}
\end{aligned}$$

Again, differentiating equation (4) with respect to θ gives us

$$\begin{aligned}
\frac{\partial}{\partial \theta}[f(z)] &= \frac{\partial}{\partial \theta}(u + iv) \\
\implies \frac{\mathrm{d}f}{\mathrm{d}z}\frac{\partial z}{\partial \theta} &= \frac{\partial u}{\partial \theta} + i\frac{\partial v}{\partial \theta} \\
\implies irf'(z)e^{i\theta} &= \frac{\partial u}{\partial \theta} + i\frac{\partial v}{\partial \theta} \quad \text{[using eqn. (3) on LHS]} \\
\implies ir\left[\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right] &= \frac{\partial u}{\partial \theta} + i\frac{\partial v}{\partial \theta} \\
\implies ir\frac{\partial u}{\partial r} + i^2 r\frac{\partial v}{\partial r} &= \frac{\partial u}{\partial \theta} + i\frac{\partial v}{\partial \theta} \\
\implies -r\frac{\partial v}{\partial r} + ir\frac{\partial u}{\partial r} &= \frac{\partial u}{\partial \theta} + i\frac{\partial v}{\partial \theta} \qquad (5)
\end{aligned}$$

Equating the imaginary and real parts on either sides of equation (5), we obtain

$$\begin{aligned}
r\frac{\partial u}{\partial r} &= \frac{\partial v}{\partial \theta} \\
\implies \frac{\partial u}{\partial r} &= \frac{1}{r}\frac{\partial v}{\partial \theta}
\end{aligned}$$

$$\begin{aligned}
-r\frac{\partial v}{\partial r} &= \frac{\partial u}{\partial \theta} \\
\implies \frac{\partial u}{\partial \theta} &= -r\frac{\partial v}{\partial r}
\end{aligned}$$

Therefore, the Cauchy-Riemann conditions in polar coordinates are

$$\boxed{\begin{aligned}
\frac{\partial u}{\partial r} &= \frac{1}{r}\frac{\partial v}{\partial \theta} \\
\frac{\partial u}{\partial \theta} &= -r\frac{\partial v}{\partial r}
\end{aligned}}$$

2 Singularities of complex functions

A point at which a complex function $f(z)$ fails to be analytic is called a *singular point* or *singularity* of $f(z)$. Various types of singularities exist, which are discussed below:

2.1 Isolated singularities

A point $z = z_0$ is called an *isolated singularity* or *isolated singular point* of a function $f(z)$ if we can find $\delta > 0$ such that the circle $|z - z_0| = \delta$ encloses no singular point other than z_0.

- If no such δ can be found, we call z_0 a *non-isolated singularity*.
- If z_0 is not a singular point and we can find $\delta > 0$ such that $|z - z_0| = \delta$ encloses no singular point, then we call z_0 an *ordinary point* of $f(z)$.

Examples

1. $f(z) = \dfrac{1}{z(z-2)}$ has isolated singularities at $z_0 = 0$ and $z_0 = 2$.
2. $f(z) = e^{1/z}$ has an isolated singularity at $z = 0$.

2.2 Poles

If z_0 is an isolated singularity and we can find a positive integer n such that

$$\lim_{z \to z_0} (z - z_0)^n f(z) \neq 0$$

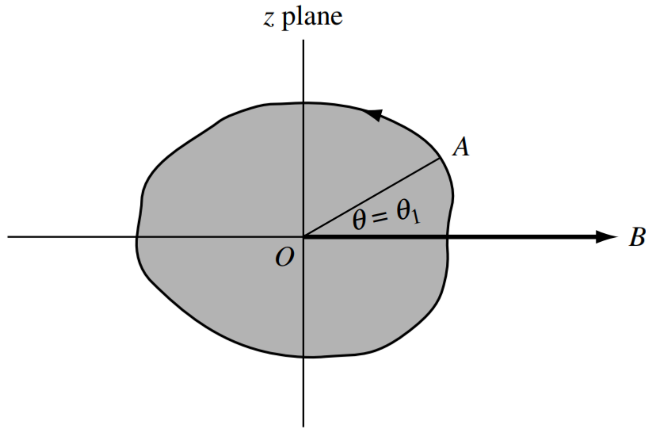
then $z = z_0$ is called a *pole* of order n.

- Special case: If $n = 1$, z_0 is called a *simple pole*.
- If $g(z) = (z - z_0)^n f(z)$, where $f(z_0) \neq 0$ and n is a positive integer, then $z = z_0$ is called a *zero* of order n of $g(z)$. In such a case, z_0 is a pole of order n of the function $\dfrac{1}{g(z)}$.
  - If $n = 1$, z_0 is called a *simple zero*.

Examples

1. $f(z) = \dfrac{1}{(z-2)^3}$ has a pole of order 3 at $z_0 = 2$.
2. $f(z) = \dfrac{3z-2}{(z-1)^2(z+5i)(z-4)}$ has a pole of order 2 at $z_0 = 1$ and simple poles at $z_0 = -5i$ and $z_0 = 4$.

2.3 Branch points



Suppose that we are given the function $w = z^{1/2}$. Further, we allow z to make a complete circuit (counterclockwise) around the origin starting from point A, as shown in the figure above.

We have $z = re^{i\theta}$, and so $w = \left(re^{i\theta}\right)^{1/2} = \sqrt{r}e^{i\theta/2}$. Then at point A, $\theta = \theta_1$ and $w = \sqrt{r}e^{i\theta_1/2}$. After a complete circuit back to A, $\theta = \theta_1 + 2\pi$ and so

$$\begin{aligned} w &= \sqrt{r}e^{i(\theta_1+2\pi)/2} \\ &= \sqrt{r}e^{i\theta_1/2}e^{i\pi} \\ \implies w &= -\sqrt{r}e^{i\theta_1/2} \end{aligned}$$

Thus, we have not achieved the same value of w with which we started. However, by making a second complete circuit back to A, i.e., $\theta = \theta_1 + 4\pi$, we have

$$\begin{aligned} w &= \sqrt{r}e^{i(\theta_1+4\pi)/2} \\ &= \sqrt{r}e^{i\theta_1/2}e^{i2\pi} \\ \implies w &= \sqrt{r}e^{i\theta_1/2} \end{aligned}$$

and we then do obtain the same value of w with which we started.

We can describe the above by stating that if $0 \leqslant \theta < 2\pi$, we are on one branch of the multi-valued function $z^{1/2}$, while if $2\pi \leqslant \theta < 4\pi$, we are on the other branch of the function.

It is clear that each branch of the function is single-valued. In order to keep the function single-valued, we set up an artificial barrier such as OB where B is at infinity which we agree not to cross.

This barrier is called a *branch line* or *branch cut*, and point O is called a *branch point*.

Examples

1. $f(z) = (z-3)^{1/2}$ has a branch point at $z_0 = 3$.
2. $f(z) = \ln\left(z^2 + z - 2\right)$ has branch points where $z^2 + z - 2 = 0$, i.e. at $z_0 = 1$ and $z_0 = -2$.

2.4 Removable singularities

An isolated singular point z0 is called a *removable singularity* of $f(z)$ if $\lim_{z \to z_0} f(z)$ exists.

By defining $f(z_0) = \lim_{z \to z_0} f(z)$, it can then be shown that $f(z)$ is not only continuous at z_0 but is also analytic at z_0.

Examples

1. $f(z) = \dfrac{\sin z}{z}$ has a removable singularity at $z_0 = 0$ because $\lim_{z \to 0} \dfrac{\sin z}{z} = 1$, which allows us to define $f(0) = 1$.
2. $f(z) = \dfrac{e^{z-1} - 1}{z - 1}$ has a removable singularity at $z_0 = 1$ because $\lim_{z \to 1} \dfrac{e^{z-1} - 1}{z - 1} = 1$, which allows us to define $f(1) = 1$.

2.5 Essential singularities

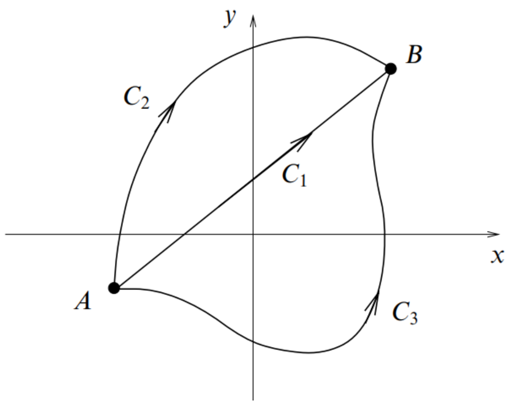
An isolated singularity that is not a pole or removable singularity is called an *essential singularity*.

Examples

1. $f(z) = e^{1/(z-2)}$ has an essential singularity at $z_0 = 2$.
2. $f(z) = \sin\left(\dfrac{1}{z - i}\right)$ has an essential singularity at $z_0 = i$.

3 Complex integration

If a complex function $f(z)$ is single-valued and continuous in some region $\mathcal{R}$ in the complex plane, then we can define the complex integral of $f(z)$ between two points A and B along some curve in $\mathcal{R}$. Its value will depend, in general, upon the path taken between A and B, as shown in the figure below.

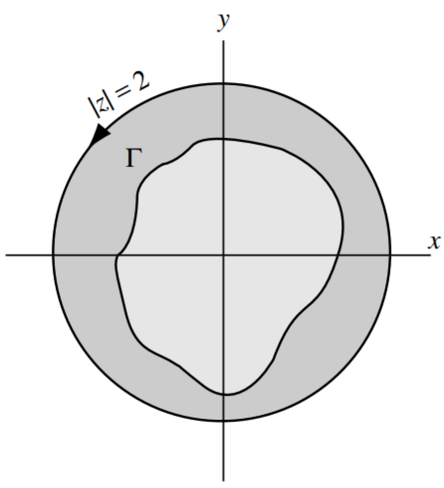


3.1 Simply and multiply connected regions

A region $\mathcal{R}$ is called *simply-connected* if any simple closed curve, which lies in $\mathcal{R}$, can be shrunk to a point without leaving $\mathcal{R}$.

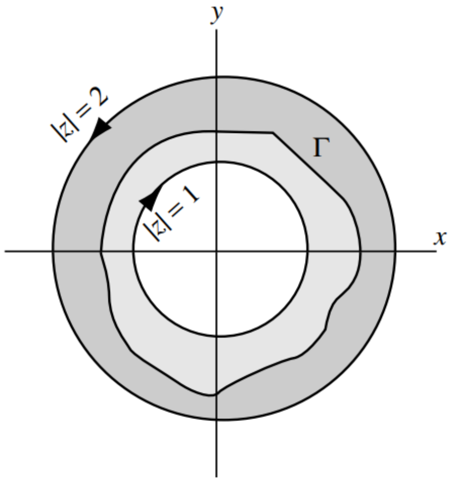
A region $\mathcal{R}$, which is not simply-connected, is called *multiply-connected*.

3.1.1 Simply connected region

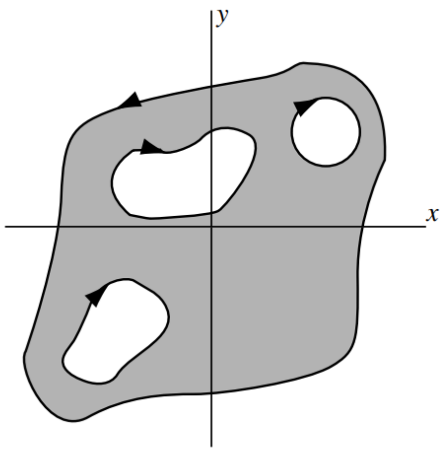


Suppose $\mathcal{R}$ is the region defined by $|z| < 2$ shown shaded in the above figure. If Γ is any simple closed curve lying in $\mathcal{R}$, i.e. whose points are in $\mathcal{R}$, we see that it can be shrunk to a point that lies in $\mathcal{R}$, and thus does not leave $\mathcal{R}$, so that $\mathcal{R}$ is simply-connected.

3.1.2 Multiply connected region



If $\mathcal{R}$ is the region defined by $1 < |z| < 2$, shown shaded in the above figure, then there is a simple closed curve Γ lying in $\mathcal{R}$ that cannot possibly be shrunk to a point without leaving $\mathcal{R}$, so that $\mathcal{R}$ is multiply-connected.



Intuitively, a simply-connected region is one that does not have any "holes" in it, while a multiply-connected region is one that does. The multiply-connected region of the above figure has three holes and, hence, is said to be triply-connected.

3.2 Green's theorem

Let $P(x, y)$ and $Q(x, y)$ be continuous and have continuous partial derivatives in a region $\mathcal{R}$ and on its boundary C. Then Green's theorem states that

$$\oint_C [P\,\mathrm{d}x + Q\,\mathrm{d}y] = \iint_{\mathcal{R}} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}x\,\mathrm{d}y$$

where $\mathcal{R}$ may be simply or multiply connected.

3.3 Cauchy's integral theorem

Let $f(z)$ be analytic in a region $\mathcal{R}$ and on its boundary C. Then

$$\boxed{\oint_C f(z)\,\mathrm{d}z = 0}$$

This fundamental theorem is known as *Cauchy's integral theorem*.

Proof

Suppose a function $f(z)$ is analytic and its derivative $f'(z)$ is continuous at all points on a simply connected region $\mathcal{R}$ which is bounded by a simple curve C.

Let,

$$f(z) = u(x,y) + iv(x,y)$$

and

$$\begin{aligned} z &= x + iy \\ \implies \mathrm{d}z &= \mathrm{d}x + i\,\mathrm{d}y \end{aligned}$$

Then we have

$$\begin{aligned} \oint_C f(z)\,\mathrm{d}z &= \oint_C [u(x,y) + iv(x,y)](\mathrm{d}x + i\,\mathrm{d}y) \\ \implies \oint_C f(z)\,\mathrm{d}z &= \oint_C (u\,\mathrm{d}x - v\,\mathrm{d}y) + i\oint_C (v\,\mathrm{d}x + u\,\mathrm{d}y) \end{aligned} \tag{1}$$

Applying Green's theorem on the first integral on the RHS of equation (1), we get

$$\oint_C (u\,\mathrm{d}x - v\,\mathrm{d}y) = \iint_{\mathcal{R}} \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) \mathrm{d}x\,\mathrm{d}y \tag{2}$$

Again, applying Green's theorem on the second integral on the RHS of equation (1), we get

$$\oint_C (v\,\mathrm{d}x + u\,\mathrm{d}y) = \iint_{\mathcal{R}} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) \mathrm{d}x\,\mathrm{d}y \tag{3}$$

Because $f(z)$ is analytic, it satisfies the Cauchy-Riemann conditions, and so we have

$$\begin{aligned} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad &\text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \\ \implies \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = 0 \quad &\text{and} \quad -\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \end{aligned}$$

Therefore, from equation (1) we obtain

$$\oint_C f(z)\,\mathrm{d}z = 0$$

which proves *Cauchy's integral theorem*.

Examples

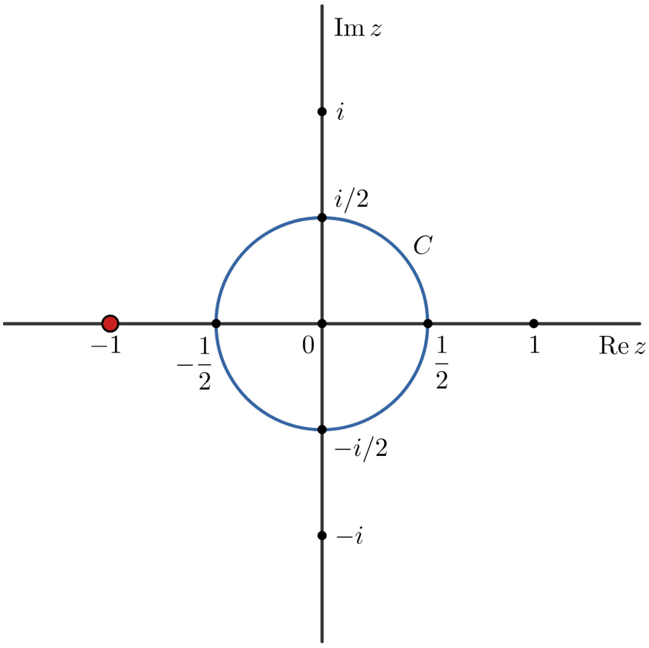
1. Evaluate the integral

$$\oint_C \frac{3z^2+7z+1}{z+1}\,\mathrm{d}z$$

where C is the circle $|z| = \frac{1}{2}$.

Solution: The integrand is $f(z) = \frac{3z^2+7z+1}{z+1}$, whose poles are obtained by setting the denominator to zero, i.e. $z+1=0 \implies z_0 = -1$.

The given circle C is centred at the origin and has a radius of $\frac{1}{2}$.



We find that the singularity $z_0 = -1$ lies outside the circle C. Hence the function $f(z)$ is analytic everywhere inside C.

Therefore, by Cauchy's integral theorem, we have

$$\oint_C f(z)\,\mathrm{d}z = 0$$

$$\implies \oint_C \frac{3z^2+7z+1}{z+1}\,\mathrm{d}z = 0$$

2. Evaluate the integral

$$\oint_C \frac{z+4}{z^2+2z+5}\,\mathrm{d}z$$

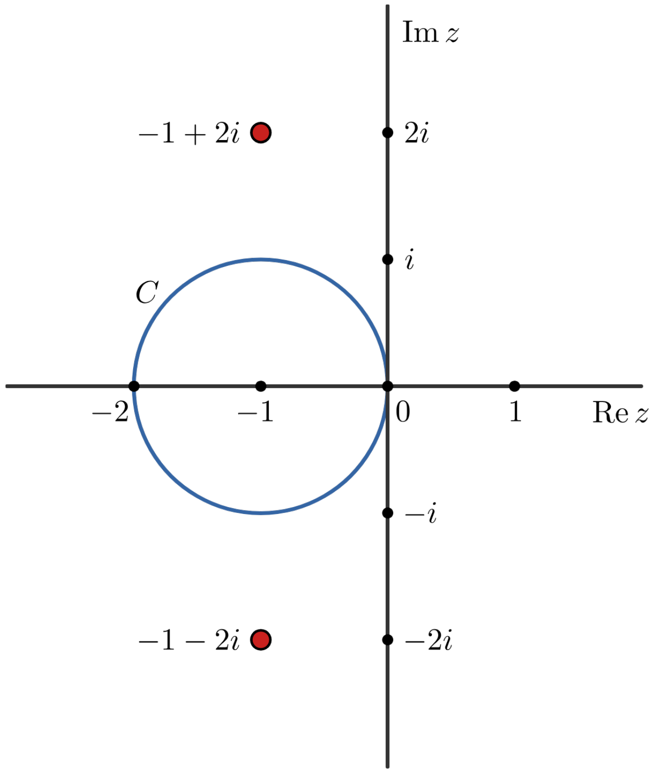
where C is the circle $|z+1| = 1$.

Solution: The integrand is $f(z) = \frac{z+4}{z^2+2z+5}$, whose poles are obtained by setting the denominator to zero, i.e.

$$\begin{aligned} & z^2+2z+5 = 0 \\ \implies & z_0 = \frac{-2 \pm \sqrt{4-20}}{2} = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2} \\ \implies & z_0 = -1 \pm 2i \end{aligned}$$

Hence the poles are $z_0 = -1 + 2i$ and $z_0 = -1 - 2i$.

The given circle C is centred at $z = -1$ and has a radius of 1.

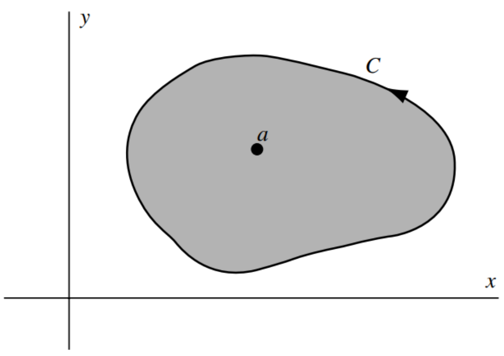


We find that both of the singularities $z_0 = -1 \pm 2i$ lie outside the circle C. Hence the function $f(z)$ is analytic everywhere inside C.

Therefore, by Cauchy's integral theorem, we have

$$\oint_C f(z)\,\mathrm{d}z = 0$$
$$\implies \oint_C \frac{z+4}{z^2+2z+5}\,\mathrm{d}z = 0$$

3.4 Cauchy's integral formulas



Let $f(z)$ be analytic inside and on a simple closed curve C and let a be any point inside C. Then

$$f(a) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-a}\,\mathrm{d}z$$

where C is traversed in the positive (counterclockwise) sense.

Also, the n^{th} derivative of $f(z)$ at $z = a$ is given by

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^{n+1}}\,\mathrm{d}z, \quad n = 1, 2, 3, \ldots$$

Examples

1. Evaluate the integral

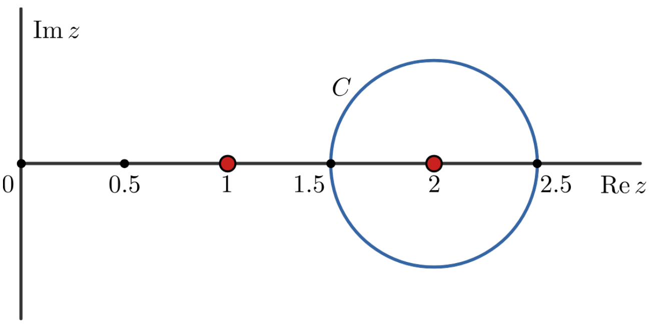
$$\oint_C \frac{z\,\mathrm{d}z}{z^2 - 3z + 2}$$

where C is the circle $|z-2| = \dfrac{1}{2}$.
Solution: We find the poles of the integrand by setting the denominator to zero and solving as follows:

$$\begin{aligned}
z^2 - 3z + 2 &= 0 \\
\implies z^2 - z - 2z + 2 &= 0 \\
\implies z(z-1) - 2(z-1) &= 0 \\
\implies (z-1)(z-2) &= 0 \\
\implies z_0 &= 1, 2
\end{aligned}$$

Hence the poles are $z_0 = 1$ and $z_0 = 2$.

The given circle C is centred at $z = 2$ and has a radius of $\dfrac{1}{2}$.



We find that that out of the two singularities, the pole at $z_0 = 2$ lies inside C.

We choose $f(z) = \dfrac{z}{z-1}$, then $f(z)$ is analytic everywhere within C. We can then write the integrand as

$$\frac{z}{z^2 - 3z + 2} = \frac{z}{(z-1)(z-2)} = \frac{f(z)}{z-2}$$

By Cauchy's integral formula, we have

$$\begin{aligned} f(2) &= \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-2}\,\mathrm{d}z \\ \implies \oint_C \frac{f(z)}{z-2}\,\mathrm{d}z &= 2\pi i f(2) \\ &= 2\pi i\left(\frac{2}{2-1}\right) \\ \implies \oint_C \frac{z\,\mathrm{d}z}{z^2-3z+2} &= 4\pi i \end{aligned}$$

2. Evaluate the integral

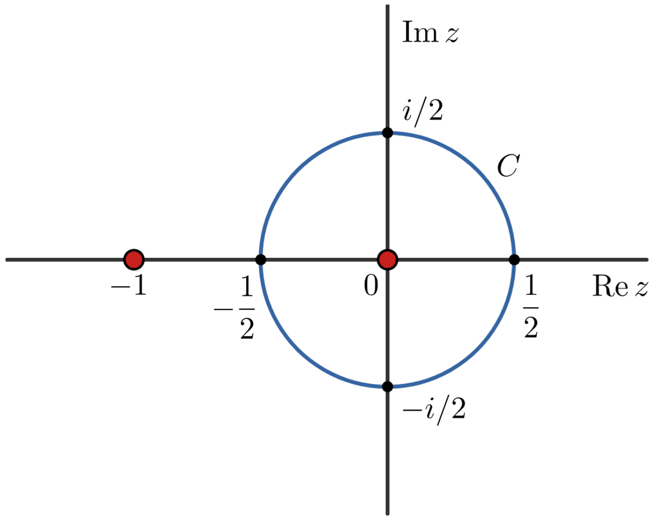
$$\oint_C \frac{2z+1}{z^2+z}\,\mathrm{d}z$$

where C is the circle $|z| = \dfrac{1}{2}$.
Solution: We find the poles of the integrand by setting the denominator to zero and solving as follows:

$$\begin{aligned} z^2+z &= 0 \\ \implies z(z+1) &= 0 \\ \implies z_0 &= 0, -1 \end{aligned}$$

Hence the poles are $z_0 = 0$ and $z_0 = -1$.

The given circle C is centred at $z = 0$ and has a radius of $\dfrac{1}{2}$.



We find that that out of the two singularities, the pole at $z_0 = 0$ is enclosed within C.

We choose $f(z) = \dfrac{2z+1}{z+1}$, then $f(z)$ is analytic everywhere within C. We can then write the integrand as

$$\frac{2z+1}{z^2+z} = \frac{2z+1}{z(z+1)} = \frac{f(z)}{z}$$

By Cauchy's integral formula, we have

$$
\begin{aligned}
f(0) &= \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-0}\,\mathrm{d}z \\
\implies \oint_C \frac{f(z)}{z}\,\mathrm{d}z &= 2\pi i f(0) \\
&= 2\pi i\left(\frac{2\times 0+1}{0+1}\right) \\
\implies \oint_C \frac{2z+1}{z^2+z}\,\mathrm{d}z &= 2\pi i
\end{aligned}
$$

4 Complex series

- *Power series*: A series in powers of $(z - z_0)$ is known as a power series, i.e.

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots + a_k(z - z_0)^k + \cdots$$

  where z is a complex variable and z_0 is the centre of the series. The constants a_0, a_1, a_2, $\ldots$ are known as the coefficients of the series.

- *Region of convergence*: The region of convergence for a power series is a set of points over which the series is defined. There may be three distinct possibilities for the region of convergence:

  i. The series converges only at the point $z = z_0$.
  ii. The series converges for all points z on the complex plane.
  iii. The series converges everywhere inside a circular region given as $|z - z_0| < R$, where z_0 is the *centre of convergence* and R is its *radius of convergence*.

- *Radius of convergence*: Consider the power series which is centred at the origin, i.e. $\sum_{n=0}^{\infty} a_n z^n$. Then its radius of convergence R is given by

  a) $\dfrac{1}{R} = \lim_{n\to\infty} |a_n|^{1/n}$

  b) $\dfrac{1}{R} = \lim_{n\to\infty} \left|\dfrac{a_{n+1}}{a_n}\right|$

Examples

1. Find the radius of convergence of the power series $1 + z + \dfrac{z^2}{2!} + \dfrac{z^3}{3!} + \cdots$.
   Solution: The given power series can be written as

$$\sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$
$$\implies a_n = \frac{1}{n!}$$

   Hence the radius of convergence is

$$\frac{1}{R} = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$
$$\implies \frac{1}{R} = \lim_{n\to\infty} \left|\frac{1/(n+1)!}{1/n!}\right| = \lim_{n\to\infty} \left|\frac{n!}{(n+1)!}\right| = \lim_{n\to\infty} \left|\frac{n!}{(n+1)n!}\right|$$
$$\implies \frac{1}{R} = \lim_{n\to\infty} \left|\frac{1}{n+1}\right| = \frac{1}{\infty}$$
$$\implies \boxed{R = \infty}$$

   which implies that the given power series converges everywhere on the complex plane.

2. Find the radius of convergence of the power series $\displaystyle\sum_{n=0}^{\infty} \frac{z^n}{2^n+3}$.

Solution: We have here

$$\begin{aligned}
\sum_{n=0}^{\infty} a_n z^n &= \sum_{n=0}^{\infty} \frac{z^n}{2^n+3} \\
\implies a_n &= \frac{1}{2^n+3} \\
\therefore\ a_{n+1} &= \frac{1}{2^{n+1}+3}
\end{aligned}$$

Hence the radius of convergence is

$$\begin{aligned}
\frac{1}{R} &= \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \\
\implies \frac{1}{R} &= \lim_{n\to\infty}\left|\frac{1/(2^{n+1}+3)}{1/(2^n+3)}\right| = \lim_{n\to\infty}\left|\frac{2^n+3}{2^{n+1}+3}\right| = \lim_{n\to\infty}\left|\frac{2^n(1+3/2^n)}{2^n(2+3/2^n)}\right| \\
\implies \frac{1}{R} &= \frac{1+3/2^{\infty}}{2+3/2^{\infty}} = \frac{1}{2} \\
\implies &\boxed{R=2}
\end{aligned}$$

which implies that the given power series converges for all points in the circular region $|z| < 2$.

3. Find the radius of convergence of the power series $\displaystyle\sum_{n=0}^{\infty} \frac{n!}{n^n} z^n$.

Solution: We have here

$$\begin{aligned}
\sum_{n=0}^{\infty} a_n z^n &= \sum_{n=0}^{\infty} \frac{n!}{n^n} z^n \\
\implies a_n &= \frac{n!}{n^n} \\
\therefore\ a_{n+1} &= \frac{(n+1)!}{(n+1)^{n+1}}
\end{aligned}$$

So we obtain

$$\begin{aligned}
\frac{a_{n+1}}{a_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \\
&= \frac{\cancel{(n+1)}\cancel{n!}}{(n+1)^n \cancel{(n+1)}} \cdot \frac{n^n}{\cancel{n!}} = \frac{n^n}{(n+1)^n} \\
&= \frac{\cancel{n^n}}{\cancel{n^n}(1+1/n)^n} \\
\implies \frac{a_{n+1}}{a_n} &= \frac{1}{\left(1+\dfrac{1}{n}\right)^n}
\end{aligned}$$

Hence the radius of convergence is

$$\begin{aligned}
\frac{1}{R} &= \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \\
&= \lim_{n\to\infty}\left|\frac{1}{\left(1+\dfrac{1}{n}\right)^n}\right| \\
&= \left|\frac{1}{\lim\limits_{n\to\infty}\left(1+\dfrac{1}{n}\right)^n}\right|
\end{aligned}$$

Now, the Euler's constant e can be obtained as a limit as given below

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$$

Therefore, we finally obtain the radius of convergence as

$$\begin{aligned}
\frac{1}{R} &= \left|\frac{1}{e}\right| = \frac{1}{e} \\
\implies \boxed{R = e}
\end{aligned}$$

which implies that the given power series converges for all points in the circular region $|z| < e$.

4.1 Taylor's series expansion of a complex function

4.1.1 Taylor's theorem

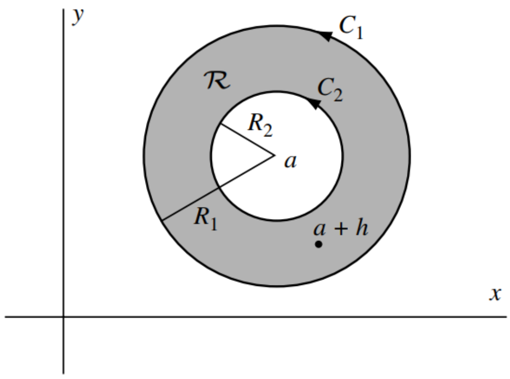
If a complex function $f(z)$ is analytic at all points inside a circle C, with its centre at the point a and radius R, then at each point z inside C, we can expand $f(z)$ in the form of a Taylor's power series as given below:

$$\begin{aligned}
f(z) &= f(a) + f'(a)(z-a) + \frac{f''(a)}{2!}(z-a)^2 + \frac{f'''(a)}{3!}(z-a)^3 + \cdots \\
\implies f(z) &= \sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(z-a)^n
\end{aligned}$$

where $f^{(n)}(a) = \left.\dfrac{\mathrm{d}^n f}{\mathrm{d}z^n}\right|_{z=a}$ is the n^{th} derivative of $f(z)$ at $z = a$.

4.2 Laurent's series expansion of a complex function

4.2.1 Laurent's theorem



Let C_1 and C_2 be concentric circles of radii R_1 and R_2, respectively, and center at a. Suppose that $f(z)$ is single-valued and analytic on C_1 and C_2 and, in the ring-shaped region $\mathcal{R}$ between C_1 and C_2. Let $z = a + h$ be any point in $\mathcal{R}$. Then we have

$$f(z) = \cdots + \frac{a_{-3}}{(z-a)^3} + \frac{a_{-2}}{(z-a)^2} + \frac{a_{-1}}{z-a} + a_0 + a_1(z-a) + a_2(z-a)^2 + a_3(z-a)^3 + \cdots$$

where

$$a_n = \frac{1}{2\pi i} \oint_{C_1} \frac{f(z)\,\mathrm{d}z}{(z-a)^{n+1}};\ n = 0, 1, 2, \ldots$$
$$a_{-n} = \frac{1}{2\pi i} \oint_{C_1} (z-a)^{n-1} f(z)\,\mathrm{d}z;\ n = 1, 2, 3, \ldots$$

with C_1 and C_2 being traversed in the positive direction with respect to their interiors.